Supplemental Material

Airway discretization in lumped simulation models of pulmonary gas exchange: how many compartments are enough?

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The verification residuals, the parameter set and the proofs referred to in the manuscript, designated as S1 to S6, are included here.

S1 Verification of the closed form

Referred to from *Closed form and periodic factor* in the manuscript's Results.

S1.1 The continuous expression against the plug-flow limit

The continuous closed form, Eq. (8) of the manuscript, should return $\min(V_D, V_T)$ as N grows. The table below evaluates it against a dead space of 150 ml.

V_T (ml)	$\min(V_D, V_T)$	closed form at $N = 10^2$	departure
500	150.00	150.0000	−0.000 %
200	150.00	149.9942	−0.004 %
165	150.00	148.6334	−0.911 %
120	120.00	119.9223	−0.065 %

Starting at $N = 10^3$ all four tidal volumes agree with the limit to four significant figures. The worst case is the tidal volume near the dead space rather than the smallest one. The source of the error lies in the portion of f_N that crosses the V_T threshold. When V_T is much greater than V_D then this tail can be ignored, and when V_T is less than V_D the limit is V_T itself, with almost all of the gas being retained in either method.

S1.2 The discrete expression against simulation

The discrete closed-form expression given as Equation (9) in the manuscript is exact and not an approximation of the scheme; the table shows how it compares with a single-pass simulation of the same scheme at tidal volumes that are whole numbers of step volumes.

N	V_T (ml)	Co	relative difference
5	300	0.10	1.7×10^{-15}
5	300	0.50	1.9×10^{-16}
40	165	0.10	3.0×10^{-15}

All three are at the level of roundoff.

S1.3 Requested against realized Courant number

If the tidal volume is not an integer number of step volumes, the step count has to be rounded off to an integer and as a result the Courant number that is actually obtained is different from the one requested. When the discrete form is evaluated at the requested value there remain residuals of order 10^{-6} to 10^{-5}, and these residuals belong to the run and not to the formula; but when it is evaluated at the actual realized value all the residuals return to machine roundoff.

N	requested Co	realized	residual (requested)	residual (realized)
5	0.10	0.1000	1.5×10^{-15}	4.5×10^{-15}
5	0.30	0.3030	6.5×10^{-5}	0
5	0.70	0.7143	6.8×10^{-5}	1.9×10^{-16}
5	0.90	0.9091	1.8×10^{-6}	1.9×10^{-16}
40	0.30	0.2993	6.8×10^{-6}	4.9×10^{-15}

Across requested values of 0.10, 0.30, 0.50, 0.70 and 0.90 at both compartment counts, every residual evaluated at the realized Courant number lies between zero and 5×10^{-15}.

When the periodic factor K is calculated with the denominator evaluated at the requested Courant number, four of the 161 configurations come out above 2, the highest being 2.0143, a result which Proposition 3 shows to be impossible. When the denominator is evaluated at the actual value however, no configuration exceeds the bound. The proof was correct and the instrument was at fault.

S1.4 The effective Courant number of a half-sine

Referred to from *The Courant number* in the manuscript's Results.

When the Courant number varies within a phase there is more than one possible average. A more appropriate question is which constant Courant number would produce the same error in the case of square flow. This value lies between 0.49 and 0.58 of the peak, being steady at the lower part of the range and then decreasing as the peak increases.

peak Co (half-sine)	equivalent constant Co	ratio to peak
0.10	0.058	0.58
0.20	0.117	0.58
0.30	0.174	0.58
0.50	0.282	0.56
0.75	0.408	0.54
0.90	0.465	0.52
0.99	0.488	0.49

When measured with $N = 5$ and $r = 0.50$, the equivalent value is obtained by means of bisection on the square-flow error. The criterion is entered at Co = 0.10, and for a half-sine this corresponds to a peak of 0.20, the first two rows of the table covering that point. The ratio is 0.58 at this location, so reducing the peak by half is conservative by the widest margin in the table.

The time average of a half-sine is $2/\pi = 0.637$ of the peak value, and the flow-weighted average is $\pi/4 = 0.785$, and neither matches. A half-sine driven to 0.99 acts like a square wave at about 0.49, which is the source of the fivefold improvement mentioned in the manuscript and also where it ceases. Since the ratio itself varies with the peak, no mean-to-peak ratio can clearly bound the achievable value.

S1.5 Proposition 3 against simulation

The argument's two steps were examined using a grid that included five different numbers of compartments, ranging from 5 to 80, and five different dead-space fractions, from 0.50 to 0.95. The deviation relative to $V_D^{\mathrm{per}} = C_e - R$ remains under 6×10^{-14}, and that relative to $V_D - C_e = R$ stays below 3×10^{-14} of V_D. Measured against R instead, the second deviation grows only where R itself approaches roundoff. Since the identity $V_D - V_D^{\mathrm{per}} = 2R$ thus holds exactly, the measured value of K agrees with $2R/\mathbb{E}[(X - V_T)^+]$ to every figure that has been reported.

The assumption is that the Courant sequences are matched in the two half-breaths; the loss of accuracy is proportional to the breaking of this assumption and the accuracy is regained when the assumption is restored. The four rows listed below all refer to the case where $N = 5$, the dead space volume is 150 ml and the ratio of inspiratory to expiratory times is 1:2, the Courant number being 0.10 at the peak of inspiration, swept over dead space fractions from 0.50 to 0.95. The departure is given as $|(V_D - V_D^{\mathrm{per}}) - 2R|$ expressed as a fraction of $V_D - V_D^{\mathrm{per}}$ and it depends on the fraction as well as on the number of compartments, which is why a range is provided rather than a single value.

Condition	departure
Square flow, constant Courant number, matched phases	below 10^{-4} %
Half-sine, fixed step, $n_E = 2n_I$	1.7 to 4.8 %
Half-sine, phases matched at $n_I = n_E$	0.02 to 0.12 %
Square flow, phases at 0.05 and 0.20 deliberately	7.1 to 20.1 %

The informative row is the third one. When a half-sine waveform is used, the two phases of a fixed-step run have a different number of steps and therefore different Courant sequences, in which case the hypothesis is invalid. If the same number of steps is given to both phases, then the sequences become identical regardless of the inspiratory to expiratory ratio since the flow is adjusted so that the same volume is delivered in each phase and hence $q_I \Delta t_I = q_E \Delta t_E = \pi V_T / 2n$. The identity then comes back to one tenth of a percent on the same waveform which had caused it to break.

The departure increases not only with the number of compartments but also with the dead-space fraction, reaching 9% when $N = 40$ and $r = 0.68$, which means that the second row represents a lower bound rather than an upper bound. In cases where the dead-space fraction is small but the number of compartments is large, the deficit for a single pass has dropped down to the level of roundoff, since the ratio is then calculated from two quantities that are negligible and therefore contains no information; these points are here excluded on the same 10^{-6} basis that was applied to K in *Closed form and periodic factor* in the manuscript.

S2 The cost of the parcel scheme

Referred to from *The chain in isolation* in the manuscript's Methods.

The parcel scheme, by virtue of how it has been constructed, is exact and the only issue concerning it is cost, the amount of which depends completely on whether the alveolar boundary is permitted to swing. The results given below correspond to a volume of 165 ml for V_T against 150 ml for V_D, giving a value of $r = 0.91$, and these were obtained over four hundred steps per phase. When the boundary is kept constant, the buffer merges almost all of the arriving parcels and on average carries 1.8 parcels with about two thousand state updates per breath. If the boundary is allowed to swing by half a millimeter of mercury, which is less than the swing in the reference model, the number rises to 378 parcels and 4.5×10^5 updates, an increase by a

factor of 210; beyond that point the buffer shows no sensitivity to the amplitude, since swings of 0.5, 2 and 4 mmHg all yield the same value when rounded to five significant figures.

The cheap figure is in fact an artefact of the test problem. For the same reason, the cost of the parcel scheme is not listed next to that of the upwind scheme in Table 3 of the manuscript. The upwind values correspond to each scheme's own cheapest configuration achieving one percent, obtained from a joint search over N and Co, while the parcel scheme's cost was determined at a fixed number of four hundred steps per phase. Putting the two entries in adjacent columns would lead to a comparison on a like-for-like basis, whereas the measurements do not support such a comparison.

S3 Alveolar ventilation cannot be read from the Bohr dead space

Referred to from *The reference model with the pattern prescribed* in the manuscript's Results.

The equation usually given in textbooks, $\dot{V}_A = f\,(V_T - V_D^{\mathrm{Bohr}})$, is based on the assumption of an airway of fixed volume. However, the reference model has a compliant airway and its volume varies during the breath, ranging from 132 to 163 ml; therefore, the equation acts as an approximation rather than representing the true value. The quality of this approximation can be tested without the need to instrument the model, since the principle of conservation means that the model must satisfy a certain identity. When the system is at steady state, the amount of carbon dioxide eliminated must equal the amount produced, and so $P_{a\mathrm{CO}_2}\dot{V}_A = 863\,\dot{V}_{\mathrm{CO}_2} = 215.75\,\mathrm{mmHg\,l\,min^{-1}}$ should be true in both arms and in every row, but the table below shows that it does not.

V_D/V_T	0.304	0.413	0.520	0.629	0.736
$N = 5$	208.23	207.63	208.15	208.77	210.04
$N = 40$	206.69	205.00	203.84	202.42	200.89

The final two rows of the apparatus sweep contain no entry since the refined arm does not have a steady state in those rows. Incomplete settling cannot account for the other cases, as each value has a residual below 0.05 mmHg. Neither arm reaches the required target on its own, the coarse arm falling short by 2.6 to 3.8 percent and the refined arm by 4.2 to 6.9, and it is only the refined arm that shows a monotonic drift during the sweep. The two arms differ from one another by 0.7 percent when the dead space is smallest and by 4.6 percent when it is largest.

The products shown in the tidal-volume sweep in Table 5 of the manuscript show different values. In a normal lung the coarse values are 207.1, 208.7, 210.3 and 212.8 whereas the refined values are 203.9, 204.6, 205.2 and 205.2, the coarse arm now rising across the sweep where it did not do so in the apparatus sweep. In the case of the injured lung the coarse values are 226.0, 229.6 and 233.9 compared with the refined values of 211.6, 211.5 and 211.4, crossing the target from below and finishing above it.

The change in sign eliminates one possible explanation. An error in units, or the omission of an STPD-to-BTPS conversion, would multiply each value by the same constant and would result in an offset of about the magnitude observed here. It is not capable of bringing the coarse arm below the target in one case and above it in another. Thus, the compliant-airway explanation passes this test whereas a units error does not.

As a result, the manuscript does not include alveolar ventilation in Tables 4 and 5. Since the conservation equation links $\dot{V}_A$ exactly to $P_{a\mathrm{CO}_2}$, any difference between the arms in the proxy is equal to the difference between the arms in the arterial tension, together with whatever error the proxy has introduced, and the tension is the value that is actually measured. In the case

of Table 4, the proxy can be obtained from the Bohr dead spaces given in that table and the tidal volume realized by each row. However, in order to settle the residual discrepancy itself, the model would have to be instrumented to record the volume entering the alveolus and the fraction of that volume which is fresh gas, and therefore the magnitude of the compliance term is limited by this check rather than determined by it.

S4 The criterion read in reverse

Referred to from *Closed form and periodic factor* in the manuscript's Results and from *Recommendations* in its Discussion.

The criterion returns a compartment count for a stated tolerance. Read the other way it returns, for a count a model already has, the largest dead-space fraction at which that count still meets the tolerance.

N	$r_{\max}$ at one percent	$r_{\max}$ at five percent
5	0.480	0.615
10	0.608	0.742
20	0.719	0.841
40	0.807	0.912

Each entry solves $2\,e_{\mathrm{disc}}(N, \mathrm{Co}, r) = \varepsilon$ for r by bisection, with e_{disc} from the discrete closed form at $\mathrm{Co} = 0.10$ against a constant alveolar boundary. The factor two is the bound of Proposition 3 and not a measured value, so the entries are conservative wherever the bound holds. Direct simulation reaches the five percent bound slightly later than the criterion does, near $r = 0.625$ against 0.615 at $N = 5$, which is the margin the factor two buys.

S4.1 Why two is a bound and 2.86 is not

It is safe to substitute $K = 2$ only in cases where the hypotheses of Proposition 3 are satisfied; in those cases where they are not, the natural question to ask is whether the largest value observed can be used instead. This is not possible.

The table given below shows the values of K for the grid of *Closed form and periodic factor* in the manuscript, with the quantity e_{disc} read at the peak Courant number under a half-sine condition with a fixed step.

N	$r = 0.50$	0.68	0.75	0.83	0.86	0.91	0.95
5	2.173	1.918	1.823	1.722	1.687	1.633	1.592
10	2.356	2.055	1.937	1.801	1.753	1.676	1.619
20	2.709	2.216	2.077	1.906	1.840	1.733	1.652
40	—	2.429	2.243	2.039	1.954	1.809	1.695
80	—	2.856	2.475	2.198	2.098	1.912	1.754

The dashes refer to points at which the single-pass deficit has dropped below 10^{-6} of the plug-flow value, the ratio being calculated from two quantities that are themselves negligible. K increases steadily with N in every column and decreases with r in every row, and there is no indication in the table that it is tending toward a limit. When the same grid is truncated at $N = 40$ the highest value obtained is 2.71 and when it is taken to $N = 80$ the result is 2.856. Any value obtained in this way is in fact the largest one found on the particular grid that was examined, and it would therefore be incorrect to quote it as a bound.

What the table in fact reflects is a discrepancy in convention. The value of e_{disc} at the peak represents the error for a chain which operates at that peak for the entire period, whereas the chain in fact spends almost the whole time below that level. The gap increases with N since the error then becomes a more steep function of Co. Reading e_{disc} at half the peak instead gives the values below.

N	$r = 0.50$	0.68	0.75	0.83	0.86	0.91	0.95
5	1.921	1.795	1.727	1.650	1.622	1.578	1.544
10	1.932	1.871	1.804	1.710	1.674	1.613	1.567
20	1.919	1.921	1.878	1.783	1.738	1.658	1.594
40	—	1.922	1.923	1.858	1.812	1.715	1.629
80	—	1.893	1.917	1.914	1.883	1.785	1.674

The maximum is 1.932 and the table now tracks the square-flow values cell by cell, which reach 2.000 at their own maximum and are the values Proposition 3 bounds. Halving the peak is slightly more conservative than the 0.49 to 0.58 that Section S1 measures for the effective ratio, and that margin is what keeps every entry below two. The remedy for a half-sine is therefore to read the Courant number in the right place, not to inflate K.

S5 Reference model parameter set

The values in the table are those the reference model runs at unless a sweep in the manuscript states otherwise. They are transcribed from the model's parameter file. The model itself is not distributed with this article, which is why the set is given here in full. With it and the structure described in the manuscript's Methods, an equivalent model can be built. Sample interval $5 \times 10^{-4}\,\mathrm{s}$, fixed-step fourth-order Runge–Kutta.

The dissociation relations are the saturation form each gas takes in the presence of the other. Writing x for the gas of interest and y for the other, content follows from the two partial pressures as

$$X_x \;=\; \frac{1 + \beta_x P_y}{K_x\,(1 + \alpha_x P_y)}\,P_x, \qquad C_x \;=\; 0.0224\,C_{\mathrm{sat},x}\,\frac{X_x^{1/h_x}}{1 + X_x^{1/h_x}},$$

with C_{sat} in $\mathrm{mmol\,l^{-1}}$ and the factor 0.0224 returning a content in ml per ml of blood. The model needs the inverse as well, and takes it by holding the other gas at its value from the previous step.

$$P_x \;=\; K_x\,\frac{1 + \alpha_x P_y}{1 + \beta_x P_y}\left(\frac{C_x}{0.0224\,C_{\mathrm{sat},x} - C_x}\right)^{h_x}.$$

The bracket is raised to a fractional power, so the expression has no real value once the content leaves $[0,\ 0.0224\,C_{\mathrm{sat},x}]$.

In the runs described in this manuscript, three values are changed, and each is specified where it appears. Carbon dioxide production is set to $250\,\mathrm{ml\,min^{-1}}$ instead of the $3.25\,\mathrm{ml\,s^{-1}}$ in the parameter table, and oxygen consumption is set to $312.5\,\mathrm{ml\,min^{-1}}$ instead of $3.75\,\mathrm{ml\,s^{-1}}$. These values result in a respiratory exchange ratio of 0.80. For the tidal-volume sweep, one block uses an alveolar compliance of $0.040\,\mathrm{l\,cmH_2O^{-1}}$, while the closed-loop sweep uses values of 0.050, 0.030, and 0.025.

It is important to make the exchange ratio clear since if carbon dioxide production is set without also setting oxygen consumption the model ends up at 1.11 instead of 0.80; in each of the runs mentioned in the manuscript both carbon dioxide production and oxygen consumption are set.

	Quantity	Value
Mechanics		
C_l, C_{tr}, C_b	larynx, trachea, bronchi ($\mathrm{l\,cmH_2O^{-1}}$)	0.00127, 0.0016, 0.0131
C_A	alveolar compliance ($\mathrm{l\,cmH_2O^{-1}}$)	0.20
C_{cw}	chest-wall compliance ($\mathrm{l\,cmH_2O^{-1}}$)	0.22
R_{ml}, R_{lt}, R_{tb}, R_{bA}	airway resistances ($\mathrm{cmH_2O\,s\,l^{-1}}$)	1.021, 0.337, 0.306, 0.082
R_{ett}	endotracheal tube resistance ($\mathrm{cmH_2O\,s\,l^{-1}}$)	4.00
V_{ul}, V_{utr}, V_{ub}	unstressed volumes (ml)	34.4, 6.63, 18.7
FRC	functional residual capacity (l)	2.4
P_{pl}	pleural pressure at end expiration ($\mathrm{cmH_2O}$)	−5.0
I:E	inspiratory to expiratory ratio	1:2
Gas exchange		
F_IO_2, F_ICO_2	inspired fractions	0.21, 0.0004
P_{atm}, P_{ws}	barometric and water vapor pressure (mmHg)	760, 47
shunt	venous admixture, as a fixed fraction	1.7 %
$\dot{V}_{\mathrm{O_2}}$, $\dot{V}_{\mathrm{CO_2}}$	metabolic rates ($\mathrm{ml\,s^{-1}}$)	3.75, 3.25
$V_{T,\mathrm{O_2}}$, $V_{T,\mathrm{CO_2}}$	tissue stores (ml)	7700, 6000
τ_{LT}, τ_{VL}	blood transport delays (s)	18.0, 10.0
Hb	hemoglobin ($\mathrm{g\,dl^{-1}}$)	15
pH, $\mathrm{HCO_3^-}$	initial acid–base	7.40, 24.0 $\mathrm{mmol\,l^{-1}}$
Dissociation relations, saturation form: C_{sat}, α, β, K, h		
	oxygen	9.0, 0.03198, 0.007175, 14.99, 0.3836
	carbon dioxide	86.0, 0.05591, 0.03255, 194.4, 1.819
Chemoreflex		
G_{pA}, G_{pf}	peripheral gains	1.310, 0.8735
τ_{pA}, τ_{pf}, D_p	peripheral time constants, delay (s)	83.0, 147.78, 7.0
G_{cA}, G_{cf}	central gains	0.484, 0.9
τ_{cA}, τ_{cf}, D_c	central time constants and delay (s)	105.0, 400.0, 8.0
f_{ac} range	afferent firing rate ($\mathrm{spikes\,s^{-1}}$)	0.835 to 12.2

When following a specified pattern the arterial carbon dioxide level is altered by approximately 0.05 mmHg. The Bohr dead spaces remain the same in all the figures given and so do the between-arm differences based on them. On the oxygen side, if oxygen is consumed more quickly the volume of the apparatus at which the refined arm reaches a negative argument in the dissociation block drops from 300 ml to 250 ml, and that is the origin of the last two rows in Table 4.

S6 Proofs

Referred to from *Unreachability of the limit under a fixed step* and *The periodic factor* in the manuscript's Theory. Equation and result numbers are those of the manuscript.

S6.1 Theorem 1

A periodic state exists and is unique. Each half-breath of Eq. (1) is affine in the state, and in the volume variable of Eq. (2) its linear part is the exponential of a constant bidiagonal generator whose rows sum to zero except the one adjacent to the Dirichlet boundary, which sums to -1. Over a phase of positive passed volume that exponential has strictly positive entries throughout the relevant triangle, so every row sum is strictly below one. Hence $\|L_I L_E\|_\infty < 1$, the breath map is a contraction in the supremum norm, and its fixed point is unique.

Proof. Let $P_A > P_I$ without loss of generality and write $Q_m = P_m - P_I$. Both half-breaths are quasi-positive. On expiration Eq. (1) reads $(V_D/N)\dot{Q}_m = |\dot{V}|(Q_{m+1} - Q_m)$ with $Q_{N+1} = P_A - P_I > 0$, and on inspiration $(V_D/N)\dot{Q}_m = \dot{V}(Q_{m-1} - Q_m)$ with $Q_0 = 0$. In either case

$Q_m = 0$ with its upwind neighbor non-negative gives $\dot{Q}_m \geq 0$. The non-negative orthant is therefore forward invariant over the whole breath, so $Q_m(t_0) \geq 0$ for all m at the start of one expiration implies the same at the start of the next, and the initial condition of the first breath can be taken in $[P_I, P_A]$ without loss of generality. Then Q_N is bounded below by the solution of $(V_D/N)\dot{Q}_N = |\dot{V}|(Q_{N+1} - Q_N)$ with $Q_N(t_0) = 0$, which is strictly positive for every $t > t_0$. Inductively Q_m is then strictly positive for every $t > t_0$ once Q_{m+1} is, since a strictly positive forcing term drives it away from zero. In particular $Q_1(t) > 0$ for every $t > t_0$, so the mouth compartment delivers gas strictly enriched relative to the inspirate over a set of positive flow-weighted measure, since $\int_{\text{exp}} |\dot{V}| \, \mathrm{d}t = V_T > 0$, and therefore $\bar{P}_E > P_I$ by a strictly positive margin. An expiratory pause at the start does not disturb this, because it carries no flow and so no weight. Since $V_D^{\text{per}} = V_T(P_A - \bar{P}_E)/(P_A - P_I)$, this gives $V_D^{\text{per}} < V_T$, which is the assertion. □

S6.2 Corollary 2

Proof. Each update is a convex combination given by $P_m^{n+1} = (1 - \text{Co}_n)P_m^n + \text{Co}_n P_{m-1}^n$ with $\text{Co}_n \in (0, 1)$, which means that the interval $[P_I, P_A]$ is invariant. A step with $\text{Co}_n = 0$ is equivalent to the identity operation, brings about no dependence and involves no flow; it therefore follows that however long an expiratory pause lasts it contributes nothing; only steps with $\text{Co}_n > 0$ are taken into account.

The count is then determined by two facts. The first is that the alveolar boundary enters compartment N and moves on to the next compartment at each step, so P_1 is unaffected by it in the first N steps and first becomes dependent on it at step index N. The second is that the explicit scheme removes, during the step from n to $n + 1$, the value of P_1^n which was held before that update, since Co_n multiplied by the compartment volume is exactly $|\dot{V}|\Delta t$. Therefore, an expiration consisting of M flux-carrying steps results in the expulsion of the values $P_1^0, \ldots, P_1^{M-1}$. When $M = N$ all the expelled values are independent of the boundary and the mixed expired pressure is therefore equal to P_I exactly, in which case the conclusion fails; but when $M \geq N + 1$ the value P_1^N is expelled and is then strictly dependent on the boundary.

The dependence is strict for each $\text{Co}_n \in (0, 1]$, which means that there is no need to exclude the case of a unit Courant number. When $\text{Co} = 1$, a phase contains $M = V_T N/V_D$ steps, and hence the condition $M \geq N + 1$ implies that $V_T > V_D$, and it is this fact which keeps the volume-matched schedule mentioned in Remark 1 outside the scope of the corollary in which that schedule reaches the limit. The boundary-dependent value that has been expelled enters the flow-weighted average of P_1 with a strictly positive weight, since the requirement that $\text{Co}_n > 0$ implies that $|\dot{V}| > 0$, and so the average is greater than P_I, the conclusion thus being reached as in Theorem 1. □

There are two cases that do not satisfy the corollary, and in each case no conclusions are drawn. The first of these is a chain which has been refined sufficiently, or in which the tidal volume is small enough, for an expiration no longer to contain $N + 1$ flux-carrying steps. The second case is the volume-matched schedule of Remark 1 at $V_T \leq V_D$, which runs at $\text{Co} = 1$ and does reach the limit. With this schedule, the limit is attained precisely when the tidal volume is a whole number of compartment volumes. When it is not, the partial step which concludes each phase leaves a residue, and at $N = 5$ with $V_D = 150\,\text{ml}$ the reported alveolar volume is 0 at $V_T = 30\,\text{ml}$, 1.7 ml at 45 ml, 2.5 ml at 100 ml and 5.0 ml at 140 ml.

S6.3 Proposition 3

Proof. Since the gas that is inspired contains no carbon dioxide, during one breath the chain has only one source, which is the alveolar boundary during of expiration, and two sinks, namely

the mouth at the time of expiration and the return to the alveolus during inspiration. In the case of periodic steady state the chain ends up in its original state, so $V_T P_A = V_T \bar{P}_E + Q$ where Q is the amount returned. Dividing both sides by $P_A - P_I$ yields $V_D^{\mathrm{per}} = Q$, and Q is the amount lost during the inspiration phase, that is, $C_e - R$. This step relies solely on the principle of conservation.

For the second statement, the state in the fresh-gas fraction is given by $\phi \in (0,1)^N$. During inspiration the value one is advected inward from the mouth, and during expiration the value zero is advected inward from the alveolus; thus, with σ denoting the reversal of the compartment order and κ denoting complementation, the two half-breath maps satisfy $I = SES$ for $S = \sigma\kappa$, $S^2 = \mathrm{id}$. The conjugacy holds step by step, with $SE_nS = I_n$ holding at a common Courant number, and composition preserves order, which is why the hypothesis requires the two sequences to agree term by term and not in reverse order. For the schedules employed here, a constant Courant number and a symmetrically sampled half-sine result in the sequence being palindromic and the two readings thus coinciding. Conjugacy by itself is not sufficient, since periodicity implies that $\phi^e = E\phi^i$ and $\phi^i = SES\phi^e$, which only state that $(\phi^i, S\phi^e)$ forms a two-cycle of $G = SE$. The affine map E has a substochastic linear part L, and once a half-breath includes at least N steps it follows that $\|L\|_\infty < 1$. This is due to a row sum, not a column sum. After M steps the row of L corresponding to compartment m sums to $\Pr\{S_M \le N - m\}$, where S_M counts the steps at which a given portion of gas moves forward and has a Poisson-binomial distribution with parameters $\mathrm{Co}_1, \dots, \mathrm{Co}_M$, reducing to $\mathrm{Bin}(M, \mathrm{Co})$ when the Courant number is kept constant. The relevant row is $m = 1$, and $\Pr\{S_M \le N - 1\} < 1$ precisely when $\Pr\{S_M \ge N\} > 0$, a condition which is met as soon as N of the Co_n are positive. That is the assertion that the boundary value has had time to reach the mouth. It follows that G^2 is a contraction and therefore has a unique fixed point, which implies that $\phi^i = S\phi^e$ and consequently that $V_D - C_e = R$. Adding the two statements yields Eq. (14), and dividing by $\mathbb{E}[(X - V_T)^+]$, which is the single-pass deficit since $\mathbb{E}[X] = V_D$ exactly, gives K.

No hypotheses are needed for the inequality. The residue R is the amount that remains when an inspiration of volume V_T is carried out starting from ϕ^e, whereas $\mathbb{E}[(X - V_T)^+]$ is the amount that remains when the same inspiration is carried out starting from a chain that is entirely alveolar, that is one for which $\phi \equiv 0$. Since ϕ^e is greater than or equal to zero component by component and the advection map is linear with non-negative coefficients, hence monotonic, the first of these quantities cannot exceed the second. □